

Instructions for the Final Project

Demonstrate your understanding of experimental design by conducting *the most scientific investigation of the most trivial question*. Working in small groups, you will design, run, analyze, and communicate the results of an experiment on a question of your own choosing.

Objective: Students learn the principles of experimental design and analysis by applying what they have learned in an end-to-end project, from formulating a research question, designing and executing an experiment, analyzing the data, all the way to communicating their findings.

Details: Pick a question that you and your group find genuinely interesting. The question can be about anything: cooking, study habits, productivity, exercise, games, plant growth—anything that can benefit from the scientific method. Just to give you some ideas:

- What is the ultimate recipe for Dubai chewy cookies?
- What is the best way to boil a soft-boiled egg?
- It is said that waiting 90 minutes after waking up to take your first sip of coffee is best for alertness. Is this true?

For additional inspiration, I will share a folder containing past student projects from Prof. Douglas Montgomery’s DoE course at Arizona State University.

Safety and ethics: Do not conduct experiments that pose risks to yourselves or others, and do not involve human subjects beyond your own group without checking with me first.

AI use: You are expected to disclose your use of AI in all parts of the project, following the policy in the Course Overview.

Do NOT fake experimental data. I will fail you for this.

1 Timeline

Week	Milestone	Format
6	Final Project Announcement	
9	Project Proposal Presentations	Presentations
13	Project Progress Report	Written Report
15, 16	Final Presentations	Presentations
16	Final Report	Written Report

2 Grade Breakdown

Component	Points
Final Project	35
Project Proposal Presentation	5
Project Progress Report	10
Final Presentation	5
Final Report	15

3 Groups

You will work on the Final Project in groups of **2–3 students**. There will be 6–7 groups in total.

You may

- form a group of your own choosing, OR
- wait to be assigned to a group.

If you prefer to work with a group of your own choosing, you need to register your group with me by sending an email to byolkim@sookmyung.ac.kr by **10 p.m. on Monday, April 13, 2026**. Only one student from each group needs to send the registration email; remember to include all the group members as recipients.

Students who do not register by the deadline will be assigned to a group.

4 Project Scope

Your project must satisfy one hard requirement:

Your experiment must involve at least 2 factors.

Beyond this, your eventual design should be substantial enough to demonstrate the core ideas of the course, such as **factorial structure**, **replication**, **randomization**, and **blocking**.

You do not need to commit to a specific design at the proposal stage. The proposal is about brainstorming treatments and establishing the direction of your project, and you will refine the specifics by the Project Progress Report in Week 13. As a rough guide for what “substantial” looks like:

- A 2^2 factorial with 2 replicates is on the low end.
- A 2^3 factorial with 3 replicates, or a 3^2 factorial with 2 replicates, is comfortably in range.
- A 2^4 factorial, even unreplicated, is more than enough.

A note on blocking: We will cover blocking after the proposal, but interested groups are welcome to read ahead and incorporate it into their design. Blocking is a powerful tool for controlling nuisance variation (e.g., differences among baking sessions, days, ovens, or experimenters), and a well-justified blocked design counts toward the “substantial” bar.

If you are unsure whether the idea for your experiment is likely to lead to a design that is substantial enough, err on the side of doing more rather than less. We will calibrate the scope together through

feedback, first following the Project Proposal Presentation and then following the Project Progress Report.

5 Project Proposal Presentation

The Project Proposal Presentations will take place on **Monday, May 4, 2026**.

Each group will present for **5 minutes**, followed by Q & A.

Your presentation should address the following:

1. **Research question:** What question are you trying to answer, and why is it interesting?
2. **Preliminary design details:** What treatments are you considering? What will you measure as the response? What are your experimental units?
3. **Feasibility:** What materials, equipment, and time will you need, and how will you procure them? Will you be able to fit the full experiment into the available time? What other challenges do you foresee, and how do you plan to address them?
4. **Plan:** Provide a week-by-week plan from now through the Progress Report in Week 13, including when you plan to work on which component.

After your presentation, submit your slide deck to Snowboard in lieu of a written proposal.

5.1 Evaluation Rubric

Criterion	Score
Does the project have a clear and interesting research question?	0/1
Is the proposed design substantial enough to demonstrate the core ideas of the course?	0/1
Is there a feasible plan for execution, including materials, equipment, and time?	0/1
Do the presenters communicate ideas in a sufficiently articulate manner?	0/1
Is the slide deck clear and well-organized?	0/1
Total	/5

Groups scoring below 4/5 on the proposal will be asked to meet with me separately to work out a revised plan together.

6 Project Progress Report

The Project Progress Report is due on **Tuesday, May 26, 2026**.

This is a written report in PDF that describes your experiment with enough precision that another team in the class could carry it out. It should also present quantitative evidence that the experiment is feasible.

The report should provide clear answers to the following questions.

1. **Precise definition of the treatments.**

Bad: Factor B: type of water = sugar water / vinegar water / regular tap water.

Good: Factor B: type of water –

- *Tap water:* 200 mL of tap water left at room temperature for one hour.
- *Sugar water:* 4 grams of sugar fully dissolved in 200 mL of tap water at room temperature.
- *Vinegar water:* 15 grams of vinegar added to 200 mL of tap water at room temperature.

2. Precise definition of the response variable.

Bad: Accuracy rate (number of words recalled).

Good:

- *Method:* Free recall.
- *Timing:* One week after the study session.
- *Procedure:* Participant writes down all remembered French–Korean word pairs on a blank sheet within 2 minutes.
- *Score:* Number of correctly recalled pairs. A pair counts as correct if the French word is spelled correctly (accents ignored) and the Korean translation is correct.

3. Other details of the experimental design. The list of relevant items depends on the project; below are two examples.

Bad: Keep a constant room temperature.

Good:

- *Target:* Ambient temperature between 20–22 °C.
- *Control mechanism:* Room thermostat set to 21 °C.
- *Verification:* Digital thermometer placed on the workbench, recorded before each trial.
- *Contingency:* Trial rescheduled if the reading falls outside the target range.

Bad: 20 random French words drawn from a French vocabulary book.

Good:

- *Source:* Full list of 1,500 words extracted from the vocabulary book.
- *Sampling:* In R, set `set.seed(42)` and use `sample(words, 720, replace = FALSE)`.
- *Assignment to trials:* The first 20 form list 1, the next 20 form list 2, and so on.

4. Justification for the choice of factors, levels, and responses.

Bad: Pace at 6'30", 7'00", 7'30".

Good: Pace at three levels:

- *Low:* 3 mph (walking).
- *Medium:* 7.5 mph (jogging).
- *High:* 12 mph (sprinting).

Rationale: Chosen after testing on a treadmill to verify that the agitation profile of the container differs visibly across the three levels.

5. Power analysis (correctly executed with substantiated inputs)

Worked example.

A total of 48 observations ensures a power of 0.92 at $\alpha = 0.05$, exceeding the standard level of 0.80 (assuming a minimum detectable effect of 2 points, a standard deviation of 2, Cohen's $f = 0.5$).

What this does well. The calculation itself is correctly executed: given the stated inputs ($f = 0.5$, $\alpha = 0.05$, the implied factorial design with $n = 48$), the reported power of 0.92 is the right number. The team also explicitly states the conventional threshold of 0.80 and observes that their analysis exceeds it.

What could be improved. The inputs themselves — the minimum detectable effect of 2 points, the standard deviation of 2, and consequently $f = 0.5$ — are not justified. Where do these numbers come from? Is a 2-point difference on a 1–10 scale a practically meaningful effect, or was it chosen because it makes the power look good? A complete power analysis would substantiate these inputs. The example below uses a fabricated citation purely to illustrate the format; your own power analysis should cite real sources.

According to Smith et al. (2020), a survey of single-session sleep-quality studies, the typical effect size for behavioral interventions corresponds to Cohen's $f = 0.25$. Based on our pilot ($n = 4$ nights per condition), the within-subject standard deviation is approximately $\sigma = 1.8$ on the 1–10 sleep quality scale. With these inputs, a 2^3 factorial with 3 replicates per subject ($N = 48$) achieves power of $1 - \beta = 0.72$ at $\alpha = 0.05$ for detecting a main effect of this size.

Notice that the substantiated inputs yield a less optimistic conclusion than the original (0.72 vs. 0.92). This is the kind of revelation that ought to prompt a design revision (e.g., increasing the number of replicates) rather than be glossed over.

6. Ethics or safety concerns. Address each of the following where applicable:

- Does the experiment involve human participants?
- Are there safety considerations (food allergies, physical exertion, etc.)?
- Is there a plan for obtaining consent?
- Is there a plan in case of accidents or adverse reactions?

Good example (concerns present).

- *Participants:* Human participants will be served sugar-sweetened beverages.
- *Screening:* Participants will be asked in advance about diabetes and beverage allergies and excluded if any are present.
- *Consent:* Obtained orally before tasting begins; participants may withdraw at any time.

- *Contingency:* In case of an adverse reaction, the participant will be asked to stop immediately and provided with water. The experimenters know the location of the nearest medical clinic.

Good example (no concerns). The experiment involves only the two team members memorizing French words from a textbook in a private study room. There are no external participants, no consumption, no physical risk. No specific safety or ethics measures are required.

7. **Reasonable analysis plan, including a formal model for the response.** Do you know how to analyze the type of data your experiment produces? Several of the proposed experiments will produce response types (categorical, ordinal, multivariate) that have not been covered in this course. If you are aware of this and would like to use this project to learn, that is fine, but you need to convince me that you understand the analysis method you intend to use.
8. **Modifications from the proposal.** If no modifications were requested in the feedback, state this explicitly.

Bad: We made some changes to the design as suggested in the feedback.

Good:

- *Change to response variable:* Switched from ‘volume of ice melted’ to a taste rating on a 5-point scale. *Rationale:* Volume of ice melted is determined by thermodynamics, while taste is the quantity we actually care about.
- *Recommendation not adopted:* The feedback suggested a Latin square for the order in which conditions are presented. *Reason for not adopting:* Our within-subject design with full randomization already balances order effects to first order.

9. **Clear and realistic timeline for the remaining work,** based on a sound estimate of trial duration.

Bad: Week 11: Design Finalization & Experiment Launch. Week 12: Project Progress Report. Weeks 13–14: Data Completion & Statistical Analysis. Week 15: Final Presentation & Report.

Good:

- *May 19 (early submission, 10 p.m.):* Submit progress report to instructor for early feedback.
- *May 21–25:* Continue pilot trials in case minor protocol adjustments are needed.
- *May 26:* Final progress report submitted on Snowboard.
- *May 27–28:* Prepare data-collection materials and finalize participant recruitment.
- *May 29:* Receive score; incorporate required changes if needed.
- *May 30 – June 8 (10 days):* Main data collection. Each trial takes approximately 45 minutes (15 min preparation + 20 min run + 10 min measurement + cleanup); 3 trials per day $\times$ 6 days, with 4 days of buffer for delays.
- *June 9–11:* Data entry, cleaning, exploratory analysis.

- *June 12–14*: Formal analysis and figures.
- *June 15–18*: Write final report.
- *June 19–21*: Prepare final presentation.

Length. There is no page limit, but I suggest 7–10 pages (11 pt single-spaced, 2.5-cm margins). The report should be long enough to address each question in sufficient depth without dwelling on unnecessary details.

Final report. The Progress Report is meant to serve as the first part of your Final Report. Aim for a quality level such that it can be lifted into the beginning of the Final Report with minimal modifications.

Early submission option. If you complete your progress report before the Snowboard window opens at 10 p.m. on May 19, you may submit it to me by email for early review. I will provide feedback as my schedule allows, typically within 2–3 days. The version submitted on Snowboard before the May 26 deadline is the one that will be graded; the early-submission round is an opportunity to identify and correct issues before the final submission. This is recommended for teams whose experiments have long data-collection phases.

Feedback. I will notify each team of a score without detailed feedback within 2–3 days after the deadline. Groups scoring below 8/10 are required to schedule a meeting with me as soon as possible to work out an emergency plan.

6.1 Evaluation Rubric

Criterion	Score
Are the treatments defined precisely enough to be reproducible by another team in the class?	0/1
Are the response variables defined precisely, including but not limited to instruments and measurement procedure?	0/1
Is the experimental design specified in sufficient detail, including but not limited to the randomization procedure?	0/1
Is the choice of factors, levels, and responses substantiated by pilot data or specifically cited prior studies?	0/1
Is the power analysis correctly executed, assuming its inputs? ¹	0/1
Are the inputs to the power analysis substantiated by pilot data or specifically cited prior studies?	0/1
Does the report adequately address any ethics or safety concerns raised by the study? ²	0/1
Does the report include a reasonable analysis plan, including a formal model for the response? ³	0/1
Have the modifications requested in the proposal feedback been incorporated, or are good reasons given for not following specific recommendations?	0/1
Does the report include a clear and realistic timeline for the remaining work?	0/1
Total	/10

Notes:

1. If the design turns out to be underpowered based on your analysis, you need to revise the design before submitting, not submit a report documenting an underpowered design.

2. If the study raises no reasonable ethics or safety concerns, this point is awarded by default.
3. If your response variable is of a type that has not been covered in class (e.g., categorical, ordinal, multivariate), you need to demonstrate sufficient knowledge of the analysis method you intend to use. If you cannot, you need to make changes to your design before submitting, not submit a report with a design that will produce data that you cannot analyze.

7 Final Report

Submit the following to the Snowboard Assignment with the title **Final Report** by **10 p.m. on Friday, June 19, 2026**:

1. **Report (PDF)**. No page limit. Evaluated on the completeness and quality of its content, not its length.
2. **Experimental data (CSV)**.
3. **All code to reproduce the analysis**. If you used R, submit an R script.
4. **AI chat history (PDF)**. Export and combine your sessions into a single file per the disclosure policy in the Course Overview.
5. **Contribution statement (PDF)**. A detailed description of how the work was divided among the team members.

7.1 Integrity gates

The following three items are assessed first as pass/fail gates. **A project is graded normally only if it passes all**, regardless of its merits on every other criterion.

Gate	Status
All required components are present.	Pass/Fail
There is no evidence of data tampering.	Pass/Fail
All relevant AI chat history is reported.	Pass/Fail

7.2 How the rubric is scored

Each criterion is scored on a common four-level scale:

Level	Meaning
0	Absent. Not addressed, or addressed in a way that does not meet the criterion.
1	Attempted. Addressed, but with substantial gaps or errors.
2	Adequate. Addressed to a sound standard; minor issues at most.
3	Exemplary. Addressed thoroughly and convincingly.

Criteria differ in how much they contribute to the total, indicated by a **Priority** of High, Medium, or Low. A criterion's contribution scales with both its level and its priority. The level descriptions above are deliberately general: what counts as "adequate" versus "exemplary" depends on the nature of your experiment, and will be applied in light of what your particular study makes possible.

7.3 Evaluation rubric

Criterion	Priority
1. Scientific question	
(a) Is the scientific question motivating the experiment clearly defined?	Medium
(b) Is there a convincing argument that the question is practically interesting?	Medium
(c) Is there a convincing argument that the experiment's outcome can answer the question?	High
2. Scope	
(a) Is the experimental design substantial enough to illustrate the core ideas of the course?	High
(b) Is the experimental design feasible given the constraints on scope?	Medium
3. Treatments and response	
(a) Are the treatments (including the choice of factor levels) well justified for the scientific question?	High
(b) Are the treatments defined precisely with a detailed procedure for applying each?	Medium
(c) Is the response variable well justified for the scientific question?	High
(d) Is the response defined precisely with a detailed measurement procedure?	Medium
(e) Is the unit of measurement clearly specified?	Low
4. Treatment assignment	
(a) Is the randomization procedure described completely, including the resulting treatment assignment, and correctly carried out?	High
(b) If runs were not all carried out at once, is there a timetable recorded at a resolution appropriate to the expected nuisance structure? ¹	Medium
5. Nuisance variation²	
(a) Are the sources of nuisance variation under consideration reasonably complete?	High
(b) Are the chosen strategies appropriate given the nature of the sources?	High
(c) For each controlled source, are the control measures clearly described and adequate?	Medium
(d) For each source handled by randomization, is that claim justified?	Medium
(e) If blocking is used, is the resulting blocking design clearly described and adequate?	Medium
6. Independence and confounding	
(a) Are the design and measurement procedures structured so that observations are independent?	High
(b) Does the collected data show no signs of dependence inconsistent with the assumed model?	Medium
(c) Are potential confounds among the factors of interest identified?	Medium
(d) Does the design successfully avoid or resolve confounds?	High
7. Procedure, sample size, and ethics	
(a) Are the remaining relevant details of the experimental procedure sufficiently specified?	Medium
(b) Does the sample-size analysis align with the experimental aim?	Medium
(c) Are the parameter settings in the power calculation justified?	Medium
(d) Is the power analysis correctly executed?	Medium
(e) Are any ethics or safety concerns adequately addressed? ³	Medium

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Criterion	Priority
8. Execution	
(a) Did the experiment as carried out match the planned protocol, with any deviations documented and their consequences assessed?	High
9. Analysis	
(a) Is the model correct for the experimental setup?	High
(b) Is the ANOVA table correct for the experimental setup?	High
(c) Are the model diagnostics sufficient for the experimental setup?	Medium
(d) If observations differ in time or location, is the possibility of temporal or spatial dependence checked? ⁴	Medium
(e) Are the analyses sufficiently comprehensive given the results (no reasonably expected analysis left undone)?	High
(f) Are the analyses appropriate (each has a clear rationale; none included for its own sake)?	High
(g) Are the analyses correctly executed?	High
10. Conclusions	
(a) Are the conclusions clearly stated?	Medium
(b) Are the conclusions justified by the analysis?	High
(c) If the experiment was inconclusive, is there a convincing account of why? ⁵	Medium
(d) Are limitations or directions for future research clearly identified?	Medium
(e) Are those limitations or directions reasonable?	Medium
11. Writing quality	
(a) Is the report well organized?	Low
(b) Is the report sufficiently clear? ⁶	Medium

Notes:

¹ If all runs can be carried out at once and were done so, this criterion is awarded full marks by default.

² Criterion 5(b) scores whether the *strategy* chosen for each nuisance source—control, randomization, blocking, or justifiably leaving it alone—is appropriate. Criterion 5(c–e) scores the *execution* of the chosen strategy. Here is an illustration in the case of blocking: a team with no clear reason to block that correctly refrains receives full marks on 5(b) and full marks on 5(e) by default; a team with a clear reason that blocks correctly also receives full marks on both 5(b) and 5(e); a team with a clear reason that blocks incorrectly receives full marks on 5(b), with 5(e) reduced according to the severity of the error; a team with a clear reason that does not block at all is penalized most severely on both 5(b) and 5(e); and a team that blocks without a clear reason receives a modest deduction on 5(b), with 5(e) scored on the execution of the blocking it performed.

³ If the study raises no reasonable ethics or safety concerns, this criterion is awarded full marks by default.

⁴ If the study raises no reasonable concerns about temporal or spatial dependence, this criterion is awarded full marks by default.

⁵ If the experiment is conclusive, this criterion is awarded full marks by default.

⁶ Clarity can be achieved not only through prose but through judicious use of visual aids: diagrams, photographs, tables, and plots.

7.4 Restoration of earlier points

Your Project Proposal and Project Progress Report were graded at a stage when your design was still taking shape. To reward iterative improvement, points lost on the content-related criteria of those earlier evaluations will be restored through your Final Report.

A point lost earlier is restored when *every* corresponding criterion in the Final Report rubric is addressed to an adequate standard (“adequate” or “exemplary”). Restoration is one-directional: a strong Final Report can only restore earlier points, never remove points you have already earned.

7.5 Extra credit

Criterion
(a) The report is written in clear English.
(b) The design includes a sophisticated element well motivated by the team’s problem.
(c) The analysis includes a sophisticated element well motivated by the team’s problem.
(d) The presentation includes a sophisticated element well motivated by the team’s problem.
(e) <i>Adversity credit.</i> The team overcame significant hurdles to complete the project (for example, a substantial change to the initial pitch or design).

A note on the “sophisticated element” credits. These reward design or analysis choices that are genuinely suited to your problem—particularly choices that solve a real resource or validity constraint—even when they look intimidating. They are not awarded for complexity added for its own sake. If a more sophisticated design (blocking, a Latin square, a split-plot arrangement, a more careful analysis) is the right tool for your situation, attempting it is exactly what these credits are for.

8 Final Presentation

The Final Presentations will take place on **Monday, June 15** and **Monday, June 22, 2026**. Teams present on the date they have already selected. Each presentation should aim for **15 ± 1 minutes**; the time limit does not include Q & A. As a rule of thumb, plan for roughly two minutes per slide. Unlike the Final Report, your Final Presentation must be delivered entirely in English.

Submit your slide deck to the Snowboard Assignment with the title **Final Presentation** before your presentation.

8.1 Evaluation rubric

Criterion	Priority
(a) Does the presentation run 15 ± 1 minutes?	Low
(b) Do the presenters speak at a reasonable pace and volume?	Low
(c) Is the slide deck clear and well organized?	Medium
(d) Does the presenter make a visible effort to engage the audience—speaking to the room rather than reading from a script, maintaining reasonable eye contact, and noticing and responding to audience reactions?	High

8.2 Restoration of earlier points

One aim of the Project Proposal was to give you an opportunity to practice presenting in English. To reward iterative improvement, points lost on the presentation-related criteria of the Project Proposal evaluation can be restored through your Final Presentation.

A point lost earlier is restored when *every* corresponding criterion in the Final Presentation rubric is addressed to an adequate standard (“adequate” or “exemplary”). Restoration is one-directional: a strong Final Presentation can only restore earlier points, never remove points you have already earned.

8.3 Attendance

Your own presentation. You must be present in person for your own team’s presentation. Absence results in an automatic F for the project, except in the event of a genuine emergency, which should be communicated to the instructor as soon as circumstances allow.

Your classmates’ presentations. You must also make a reasonable effort to learn about your classmates’ work. This obligation is satisfied either by attending the presentations in person or by watching the recordings, which will be posted on Snowboard within 24 hours after each presentation date. Whichever way you choose, you must have witnessed every presentation—in person, by recording, or any combination—by **the end of the day on Sunday, June 28, 2026**, after which attendance records are finalized.